

ACTIVITY 8: HACKERRANK ALGORITHMIC PROBLEM-SOLVING

PORTFOLIO BUILDING B25CS0311 • 3rd Semester B.Tech – Computer Science & Engineering

Student Name	Nandeesh
Register No.	R25EJ086
GitHub	https://github.com/nandeeshsmoolya
HackerRank	https://www.hackerrank.com/profile/nandeeshsmoolya

1. Problems and Complexity Analysis

#	Problem	Key Technique	Time	Space
1	Diagonal Difference	2D diagonal traversal	$O(N)$	$O(1)$
2	Dynamic Array	Vectors + XOR indexing	$O(N + Q)$	$O(N)$
3	Time Conversion	String parsing + logic	$O(1)$	$O(1)$
4	Compare the Triplets	Element-wise comparison	$O(1)$	$O(1)$
5	Sparse Arrays	Frequency mapping	$O(N + Q)$	$O(N)$

2. Reflective Summary

This activity strengthened my understanding of practical algorithmic problem solving and efficient implementation. Diagonal Difference helped me practice matrix traversal and calculating primary and secondary diagonal sums while using constant extra space. Dynamic Array introduced nested vectors and XOR-based indexing for processing queries. Time Conversion improved my ability to parse strings and apply AM/PM conversion rules using simple conditional logic. Compare the Triplets reinforced element-wise comparison and score tracking. Sparse Arrays showed how frequency mapping can make repeated string queries efficient instead of repeatedly scanning the complete input. Across the five problems, I learned to think about time and space complexity before choosing an implementation. I also gained experience with HackerRank's test-case based evaluation, including hidden test cases, which emphasizes writing solutions that are correct beyond the sample input. Finally, organizing the solutions in a GitHub repository helped me practice presenting code in a structured and reviewable format. Overall, the activity connected classroom concepts such as arrays, vectors, strings, hashing, and complexity analysis with hands-on coding-platform practice and portfolio building.

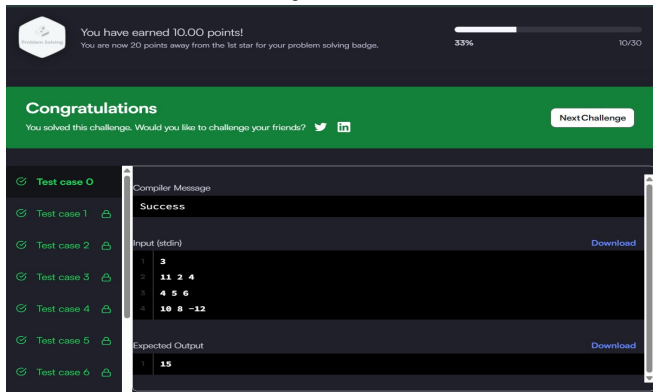
3. Repository

GitHub Repository: <https://github.com/nandeeshsmoolya/HackerRank-3rdSem-Portfolio>

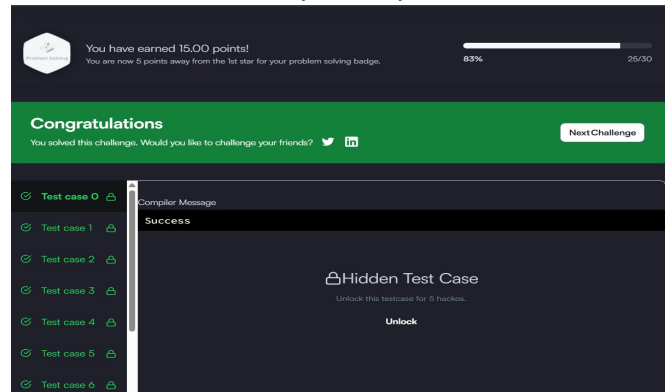
4. HackerRank Accepted Submission Evidence

The following screenshots document successful HackerRank submissions in the order of the five mandatory problems.

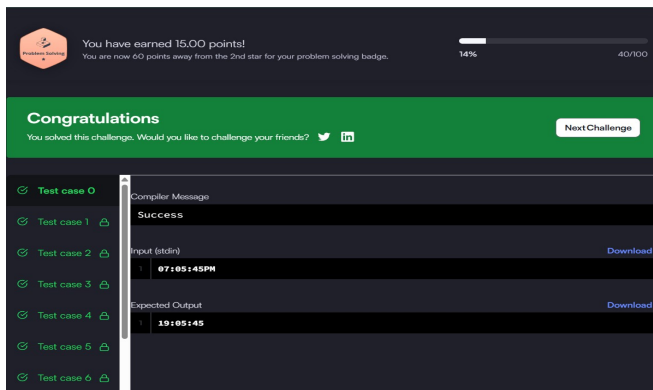
1. Diagonal Difference



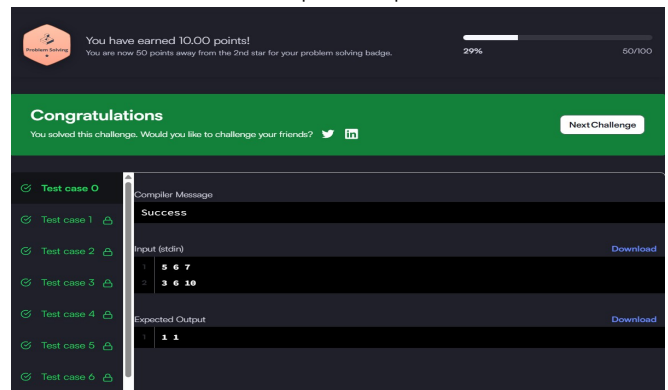
2. Dynamic Array



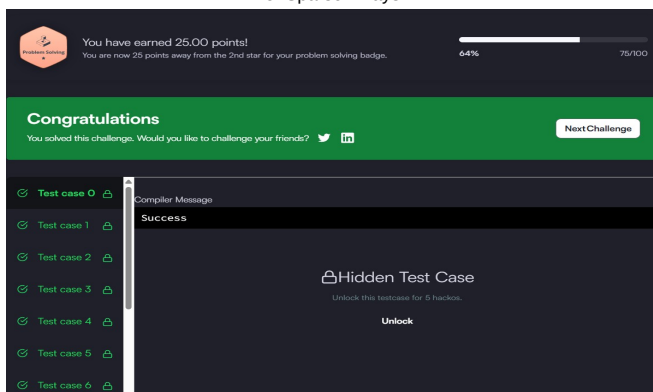
3. Time Conversion



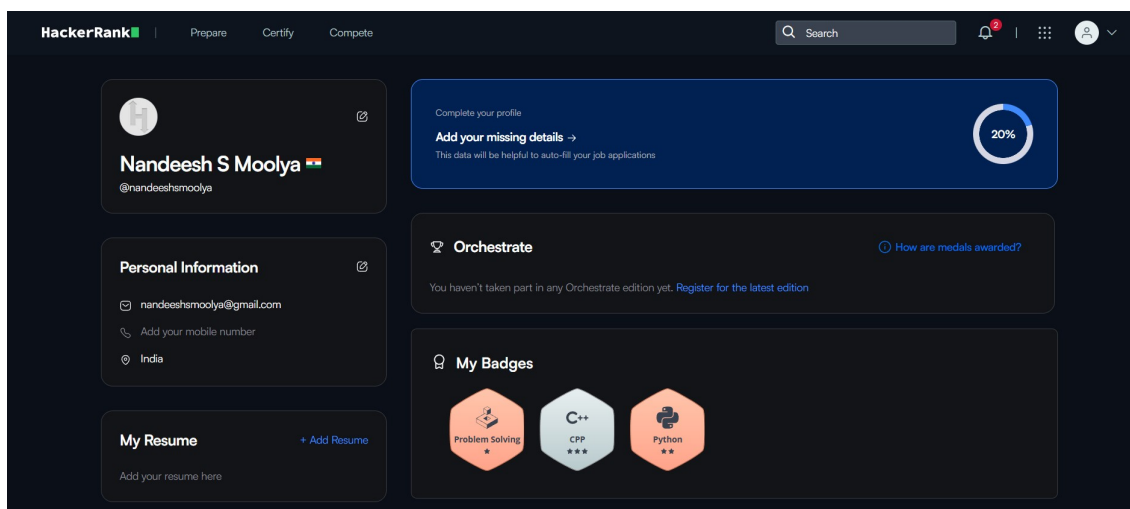
4. Compare the Triplets



5. Sparse Arrays



5. HackerRank Profile & Badge Evidence



HackerRank: <https://www.hackerrank.com/profile/nandeeshsmoolya>

GitHub: <https://github.com/nandeeshsmoolya/HackerRank-3rdSem-Portfolio>

The profile screenshot shows the C++ *** badge required by the activity.